

Your files. Your drive. Your cloud.

A professionally hosted, single-purpose file sync & share platform.
Turn a spare external HDD/SSD on an always-on host into a private cloud — like Dropbox, but fully under your control.

Server NestJS 20 + Postgres 15	Desktop Tauri • Tray • Overlay	Mobile Expo Bare • Offline	TLS Caddy • Let's Encrypt
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BLUE = Do this • AMBER = Attention • GREEN = Done • Every screen shows: ● **Status** | **Storage used/total** | **Last synced**

How it fits: Desktop ~/PersonalCloud (Tauri notify) + Mobile browse-cloud ⇕ HTTPS chunked 10 MB • ETag • SHA-256 → Server NestJS —StorageAdapter→ /mnt/storage/asm-cloud-data/{userId}/... + .versions/ + .asm-chunks/ • Postgres • Caddy :443 • WebSocket /sync push Share /api/public/share/:token (password/expiry, read-only) • Trash 30d • Versioning 5 • Conflict → "(conflicted copy YYYY-MM-DD).ext" never overwrites

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How to use: Follow **A** → **B** → **C** → **D** in order. Copy-paste the blue code blocks exactly. The drive is the volume, not the server.

01 What You Need — Shopping List

Hardware & software for a private home cloud

Item	Minimum	Tip
Always-on Host	Old laptop / mini-PC / Pi 4/5 (2 GB) / NAS	Pi 5 + SSD silent ~5 W. Don't use daily laptop.
External HDD/SSD	500 GB–4 TB USB, ext4/NTFS	This IS your cloud — back it up.
Network	Home router + optional domain	Without domain: http://192.168.1.50:3000 at home.
OS	Debian/Ubuntu, Pi OS, or Windows 10/11	Docker on Linux is one-command prod
Software	Docker + Compose (prod) or Node 20 + PG 15	DB_TYPE=sqlite dev only

Host + drive on UPS if possible — prevents DB corruption on outage.

02 How It Fits Together — Architecture

Lean Nextcloud: only file sync, no Calendar/Mail bloat

Layer	Technology	Mirrors Nextcloud	Diverges (lean)
Server	NestJS 20, Postgres 15, LocalDiskAdapter	Files + sharing + WebSocket push	NEW protocol (not WebDAV)
Desktop	Tauri 1.6 Rust+Web, tray, notify	C++/Qt tray, overlay, chunked NG	Lighter, shared TS engine
Mobile	Expo Bare, media-library/background-fetch	Kotlin/Swift auto-upload	Same engine, bare for bg
Protocol	/api/files, /sync/changes, WS /sync	PROPFIND + sync tokens	Local disk, S3 later

Every file: ETag + SHA-256, quota enforced, versioning FILE_VERSIONS_KEEP=5 (.versions/), trash 30d. Lock: never auto-overwrites → (conflicted copy).

03 Step A — Prepare the External Drive

The drive is the cloud volume • 5 minutes

Windows (dev): Plug → Format NTFS → note F: → create F:\asm-cloud-data → in server/.env set STORAGE_ROOT=F:/asm-cloud-data

Linux / Pi / NAS (prod):

```
sudo mkdir -p /mnt/storage/asm-cloud-data
sudo mkfs.ext4 /dev/sda1 # only if new/empty! wipes!
sudo mount /dev/sda1 /mnt/storage/asm-cloud-data
echo '/dev/sda1 /mnt/storage/asm-cloud-data ext4 defaults,nofail 0 2' | sudo tee -a /etc/fstab
sudo chown -R $USER:$USER /mnt/storage/asm-cloud-data
ls /mnt/storage/asm-cloud-data # empty = done
```

04 Step B — Install the Server

Pick prod (one-command) or dev (Windows)

B1 — One-command prod (Recommended, Docker + Caddy HTTPS)

```
git clone https://github.com/YOURNAME/ASM_Cloud.git
cd ASM_Cloud/server
cp .env.example .env
nano .env # CHANGE: DB_PASSWORD, JWT_*_SECRET (32+ chars), ADMIN_INVITE_TOKEN,
# STORAGE_ROOT=/mnt/storage/asm-cloud-data, PUBLIC_BASE_URL=https://cloud.example.com
```

```
nano Caddyfile # top line = your domain
cloud.example.com {
reverse_proxy server:3000
}
```

```
docker compose -f docker-compose.yml -f docker-compose.prod.yml up --build -d
docker compose logs -f server # wait "ASM_Cloud server listening on ...:3000"
```

Check: curl -X POST http://localhost:3000/api/auth/register ... → {"user":{"id":...}}

B2 — Dev on Windows (no Docker)

```
cd F:\ASM_Cloud\server
copy .env.example .env # set STORAGE_ROOT=F:\ASM_Cloud\data/storage and DB_TYPE=sqlite
npm install
npm run start:dev # http://localhost:3000/api
```

B2 DB_TYPE=sqlite is dev only. Prod uses Postgres.

Verify

```
curl -X POST http://localhost:3000/api/auth/login -H "Content-Type: application/json" -d
'{"username":"alice","password":"S3cure!pass"}'
curl -X POST http://localhost:3000/api/files/upload -H "Authorization: Bearer <token>" -F "file=@hello.txt" -F
"path=docs/hello.txt"
curl http://localhost:3000/api/files/download?path=docs/hello.txt -H "Authorization: Bearer <token>" -o hello.txt
```

05 Step C — Desktop App (Tauri)

Windows / macOS / Linux • lighter than Electron • tray + overlay

```
cd desktop
npm install
npm run dev # web preview mock http://localhost:1420
cargo install tauri-cli # needs Rust https://rustup.rs
npm run tauri:dev # real window + tray + ~/PersonalCloud picker
npm run tauri:build # installer src-tauri/target/release/bundle/
```

- 1) **Login** → Server URL http://localhost:3000 or https://cloud.example.com → alice / S3cure!pass
- 2) **Settings** → **Sync folder** → Browse... → C:\Users\You\PersonalCloud
- 3) **Selective Sync** → check docs/, Photos/ (empty = all)
- 4) **Sync now** → Activity Log: Downloaded docs/hello.txt (17 B) • Tray synced/syncing/error

Drop file into ~/PersonalCloud/docs/test.txt → tray syncing → synced. Edit same file on two devices → test (conflicted copy 2026-09-08).txt keeps both (lock).

06 Step D — Mobile App (Expo Bare)

Android / iOS • browse-cloud, not mirror

```
cd mobile
npm install --legacy-peer-deps --ignore-scripts
npx expo start # web preview
npx expo run:android # emulator needs 10.0.2.2:3000 (not localhost)
npx expo run:ios
```

Step	Action	Note
1. Login	Server URL http://192.168.1.50:3000 or HTTPS	SecureStore on device
2. Browse cloud	Root → docs/ → shared.txt → Open (206 Range) or Keep offline	Offline tab plane-readable
3. Camera	Enable auto-upload (Wi-Fi only ON) → Photos/Camera → Run now / background (15 min)	iOS 15 min, Android fg service
4. Share	From any app → Share → ASM_Cloud → Shared/	intentFilters bare rebuild

07 Everyday Use

The three clients talk to the same account

I want to...	Do this	Status
Drop file on desktop	Drag into ~/PersonalCloud/docs/	synced
Get file on phone	Browse → Open (on-demand)	206 Range
Keep for plane	Keep offline → Offline tab	offline ✓
Camera auto	Camera tab → Enable → Run now	Wi-Fi
Recover old version	GET /api/files/versions?path=docs/report.pdf → POST /versions/restore?path=...&version;=1	keep 5
Trash 30d	DELETE /api/files?path=docs/old.txt → GET /api/files/trash	soft delete
Pause sync	Desktop Pause → tray paused	gray

Every screen always shows: ● Sync status | Storage used/total | Last synced (accent #2E6FE0). Badges: synced green, syncing amber, conflict red, offline gray.

08 Admin — Users, Quotas, Storage

First registered user is isAdmin:true

```
TOKEN=$(curl -s -X POST https://cloud.example.com/api/auth/login -H "Content-Type: application/json" -d '{"username":"alice","password":"S3cure!pass"}' | jq -r .accessToken)
curl https://cloud.example.com/api/admin/users -H "Authorization: Bearer $TOKEN"
curl -X POST https://cloud.example.com/api/admin/users -H "Authorization: Bearer $TOKEN" -d '{"username":"bob","email":"bob@home","password":"Another!pass","quotaBytes":"5368709120"}'
curl -X POST https://cloud.example.com/api/admin/users/<id>/quota -H "Authorization: Bearer $TOKEN" -d '{"quotaBytes":"5368709120"}' # 5 GB
curl https://cloud.example.com/api/admin/dashboard -H "Authorization: Bearer $TOKEN"
```

Desktop AdminPanel (desktop/src/components/AdminPanel.tsx) shows drive total/free, per-user bar, Edit quota, Create user — admin only, else 403.

Quota exceeded 413 on POST /api/files/upload / chunked complete → Activity Log error.

09 Share Links & Security

File + folder, password, expiry — public read-only like Nextcloud

```
curl -X POST https://cloud.example.com/api/shares -H "Authorization: Bearer $TOKEN" -d '{"path":"docs/report.pdf","password":"s3cret","expiresAt":"2026-12-31T00:00:00Z"}'
# → {"token":"xfqRhYXQjxk_","url":"/api/public/share/xfqRhYXQjxk_"}
curl https://cloud.example.com/api/public/share/xfqRhYXQjxk_ # info (hasPassword?)
curl https://cloud.example.com/api/public/share/xfqRhYXQjxk_/download -H "X-Share-Password: s3cret" -o report.pdf
```

Desktop: SERVER FILES → Share (ShareDialog) password/expiry → copies https://.../api/public/share/.../download. Mobile: per file Share. Security: JWT 15m + refresh 7d, argon2, share argon2 + expiry 403, every endpoint checks owner/share, ETag + SHA-256, TLS via Caddy Let's Encrypt.

10 Go Live — HTTPS via Caddy

Automatic Let's Encrypt • one domain

Step	Action
1. DNS	Domain cloud.example.com → host public IP (router forward 80,443 → host)
2. Env & Caddyfile	.env PUBLIC_BASE_URL=https://cloud.example.com CORS_ORIGIN=... • Caddyfile line 1 = your domain → reverse_proxy server:3000
3. Launch	docker compose -f docker-compose.yml -f docker-compose.prod.yml up --build -d — Caddy auto-TLS + WebSocket proxy
Home-only	Skip Caddy, use http://192.168.1.50:3000

Never expose Postgres 5432 publicly — prod binds 127.0.0.1:5432:5432 only.

11 Troubleshooting — For Laymen

Copy the fix exactly

Symptom	Fix
Quota exceeded 413	Admin increase quota or POST /api/files/trash/purge — check GET /api/admin/storage
error ETag mismatch	Check (conflicted copy), pick one
offline connect_error	docker compose logs server → docker compose restart server; check ls /mnt/storage/asm-cloud-data
Mobile media permission	Settings → Apps → ASM_Cloud → Photos → Allow
Camera skip not Wi-Fi	Expected if Wi-Fi-only ON; connect Wi-Fi or OFF toggle
Drive full	GET /api/admin/storage → add HDD, rsync -a
WebSocket missing token	Login again — access token 15m expires, auto-refresh via refreshToken

Logs: docker compose logs -f server/caddy • Desktop Activity Log first. Server push RealtimeGateway → sync:changes.

12 Cheat-Sheet — All Commands

Copy-paste ready • Phase 1–8

```
# Server health
docker compose ps
docker compose logs -f server | grep ASM_Cloud
# Auth
curl -X POST http://localhost:3000/api/auth/register -H "Content-Type: application/json" -d '{"username":"alice","email":"alice@home","password":"S3cure!pass","inviteToken":"..."}'
curl -X POST http://localhost:3000/api/auth/login -H "Content-Type: application/json" -d '{"username":"alice","password":"S3cure!pass"}'
# Files
curl -X POST http://localhost:3000/api/files/upload -H "Authorization: Bearer $TOKEN" -F "file=@big.mp4" -F "path=videos/big.mp4"
# chunked auto >10MB
curl http://localhost:3000/api/files/download?path=videos/big.mp4 -H "Authorization: Bearer $TOKEN" --range 0-10485759 -o part.bin
# Shares
curl -X POST http://localhost:3000/api/shares -H "Authorization: Bearer $TOKEN" -d '{"path":"docs/report.pdf","password":"x"}'
# Admin
curl http://localhost:3000/api/admin/dashboard -H "Authorization: Bearer $TOKEN"
# Clients
cd desktop && npm run dev # or npm run tauri:dev
cd mobile && npx expo start # 10.0.2.2:3000 on Android emulator
```

Nextcloud alignment: Server PHP/WebDAV vs ASM_Cloud NEW protocol + StorageAdapter; Desktop C++/Qt vs Tauri Rust+Web (lighter); Mobile Kotlin/Swift vs Expo bare — all share packages/sync-engine chunking/diff/backoff.